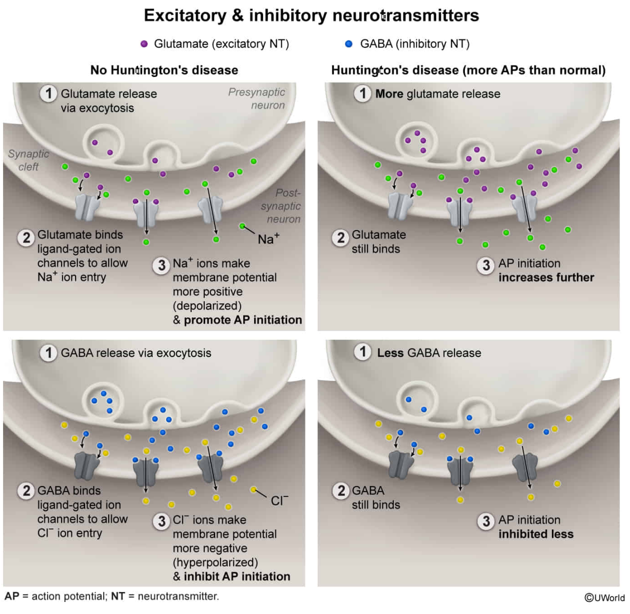
Huntington's disease is a genetic neurodegenerative disorder that causes brain neurons to experience more action potentials than normal. It is known that activity of the neurotransmitter glutamate increases in these patients and activity of the neurotransmitter GABA decreases. Given this, these two neurotransmitters most likely have which of the following effects in the brain?

- A. Glutamate is an inhibitory neurotransmitter; GABA is an excitatory neurotransmitter. [2%]
- B. Glutamate is an excitatory neurotransmitter; GABA is an inhibitory neurotransmitter. [96%]
- C. Both glutamate and GABA are inhibitory neurotransmitters.
- D. Both glutamate and GABA are excitatory neurotransmitters.

Correct
95% answered correctly

**Explanation:**




Neurons transport signals to other neurons via action potentials, the electrochemical signals used by the nervous system for communication. In presynaptic neurons (the neuron sending the signal), these action potentials travel down axons until they reach the axon terminals, which are responsible for signal transmission across the synapse. Specifically, at chemical synapses, the action potential causes the release of neurotransmitters from the presynaptic neuron into the synaptic cleft (the space between the presynaptic neuron and the postsynaptic dendrite).

Neurotransmitters bind receptors on the membrane of the postsynaptic neuron and can cause an action potential only if the membrane potential of the postsynaptic neuron becomes more positive (depolarized) and exceeds a certain threshold.  Depending on their effect on the postsynaptic cell, neurotransmitters are classified as excitatory or inhibitory.  Excitatory neurotransmitters *depolarize* the postsynaptic membrane, *promoting* an action potential.  Inhibitory neurotransmitters *hyperpolarize* the postsynaptic membrane (ie, cause the membrane to become more *negative*), *inhibiting* an action potential.

In this scenario, Huntington's disease is a neurodegenerative disorder that causes brain neurons to experience *more* action potentials than normal.  In other words, these affected neurons become depolarized more often, meaning these neurons must have *more* excitatory neurotransmitters and *fewer* inhibitory neurotransmitters acting upon them.

The question states that in these patients, the activity of the neurotransmitter glutamate increases, and the activity of the neurotransmitter GABA decreases. Because excitatory signals increase in these patients, **glutamate** is likely an **excitatory** neurotransmitter.  Conversely, because inhibitory signals decrease in these patients, **GABA** is likely an **inhibitory** neurotransmitter **(Choices A, C, and D)**.

**Educational objective:**
Excitatory neurotransmitters depolarize postsynaptic membranes, increasing the likelihood of action potential initiation.  Inhibitory neurotransmitters hyperpolarize postsynaptic membranes, decreasing the likelihood of action potential initiation.

**Time spent:**0 sec
**Subject:**
Biology

**QID**:403112
**System:**
Endocrine and Nervous Systems